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EEE 533 — Past Questions & Materials

by Renance

Course: EEE 533

Institution: Federal University of Petroleum Resources, Effurun

Papers in this pack: 1

Levels: 500

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Paper 1 of 1 — EEE 533 DIGITAL SIGNAL PROCESSING 2019,2020

Level 500 • Second Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN
COLLEGE OF TECHNOLOGY

EXAMINATION: Second Semester

SESSION: 2019/2020

DEGREE: B. Eng.



COURSE CODE: EEE 533

COURSE TITLE: Digital Signal Processing

TIME ALLOWED: $2\frac{1}{2}$ Hours

DATE: July 23, 2021

Instructions: Answer any four Questions.

Question One (17.5marks)

(a) Briefly define the following terms as used in digital signal processing:
(i) Signals (ii) Stochastic Signals (iii) Energy Signals (iv) Power signals (v) Causal signals. (2.5 marks)

(b) Briefly highlight the basic elements of digital signal processing (2.5 marks)

(c) List four advantages of digital signal processing over Analogue signal processing (2 marks)

(d) Determine and classify each of the following functions as either Energy or Power signal:

i. $x(t) = \begin{cases} 4(t) & t \geq \infty \\ 0 & \text{otherwise} \end{cases}$

ii. $x(t) = \begin{cases} e^{-2t} & t \geq \infty \\ 0 & \text{otherwise} \end{cases}$

10 marks

(Total marks = 17.5)

Question Two (17.5 marks)

(a) Briefly discuss the computational advantage of FFT over DFT algorithm. (3.5 Marks)

(b)i. Determine the discrete Fourier transform (DFT) of the four 4-point sequence: $x(n) = \{1, 1, 1\}$ and sketch the magnitude plot. 10 marks

(b) ii. Find the inverse DFT of the 4-point sequence $x(k) = \{1 - j, 0, 2\}$ 4 marks

(c) Determine the 4-point sequence $x(n) = \{1, 1, 1\}$ using decimation in frequency - fast Fourier transform (DIF-FFT) technique . 4 marks

Question Three (17.5 marks)

(a) What you understand by the terms canonic and non-canonic structures in the realization of Infinite Impulse Response (IIR) filters. (3.5 Marks)

(b) What is basic advantage direct form-II realization of digital filters have over direct form-I. (2 marks)

(c) Develop direct form-II realization of the transfer function given as

$$H(z) = \frac{3+3.6Z^{-1}+0.6Z^{-2}}{1+0.1Z^{-1}-0.2Z^{-2}} \quad (6 \text{ marks})$$

(d) Determine the cascade realization of a system characterized by the transfer expressed as $H(z) = \frac{2(z+2)}{z(z-0.1)(z+0.5)(z+0.4)}$ (6 marks)

Number Four (17.5 marks)

- (a) What do you understand by linear phase response of finite impulse filter? (4 marks)
 (b) Write the difference equation for FIR filter and explain the meaning of each (3.5 marks)

(c) Design a symmetric FIR low-pass filter for which the desired frequency response is expressed as

$$H_d(\omega) = e^{-j\tau} \quad \text{for } -\omega_c \leq \omega \leq \omega_c$$

$$= 0 \quad \text{elsewhere,}$$

given that the length of the filter should be 7 and $\omega_c = 1 \text{ radian/second}$. The phase delay $\tau = \frac{M-1}{2}$. Use Hamming window whose function is given

$$W_H(n) = \begin{cases} 0.54 - 0.46 \cos \frac{2\pi n}{M-1} & \text{for } n = 0, 1, \dots, M-1 \\ 0, & \text{elsewhere} \end{cases} \quad (10 \text{ marks})$$

Question Five (17.5 marks)

- (a) Draw the block diagram showing the elements that make up an image processing. Explain the function of each element. (4 marks)
 (b) What do you understand by aliasing in the context of image sampling? Explain. (3 marks)
 (c) Explain what you understand by image format. Mention any three image formats in use. (3 marks)
 (d) Discuss five areas of application of image processing. (2.5 marks)
 (e) Explain what you understand by i. Binary image ii. Gray scale image iii. Colour image. (3 marks)
 (f) What is the purpose of image enhancement in basic digital signal processing of images? (2 marks)

Question Six (17.5 marks)

- (a) Write down the difference equations for Infinite Impulse Response (IIR) filters. Explain each term clearly. (3.5 marks)
 (b) What is the basic difference between IIR and FIR filters. (2 marks)
 (c) If $H_a(s) = \frac{10}{(s+2)(s+4)}$. Is the transfer function of an analogue filter, find the corresponding $H(z)$ using impulse invariant method for a sampling frequency of 5 Hz. (6 marks)
 (d) The transfer function of an analogue filter is $H_a(s) = \frac{3}{(s+2)(s+3)}$ with period $T = 0.1 \text{ seconds}$. Design a corresponding digital impulse response filter (IIR) using Bilinear transformation method. (6 marks)